

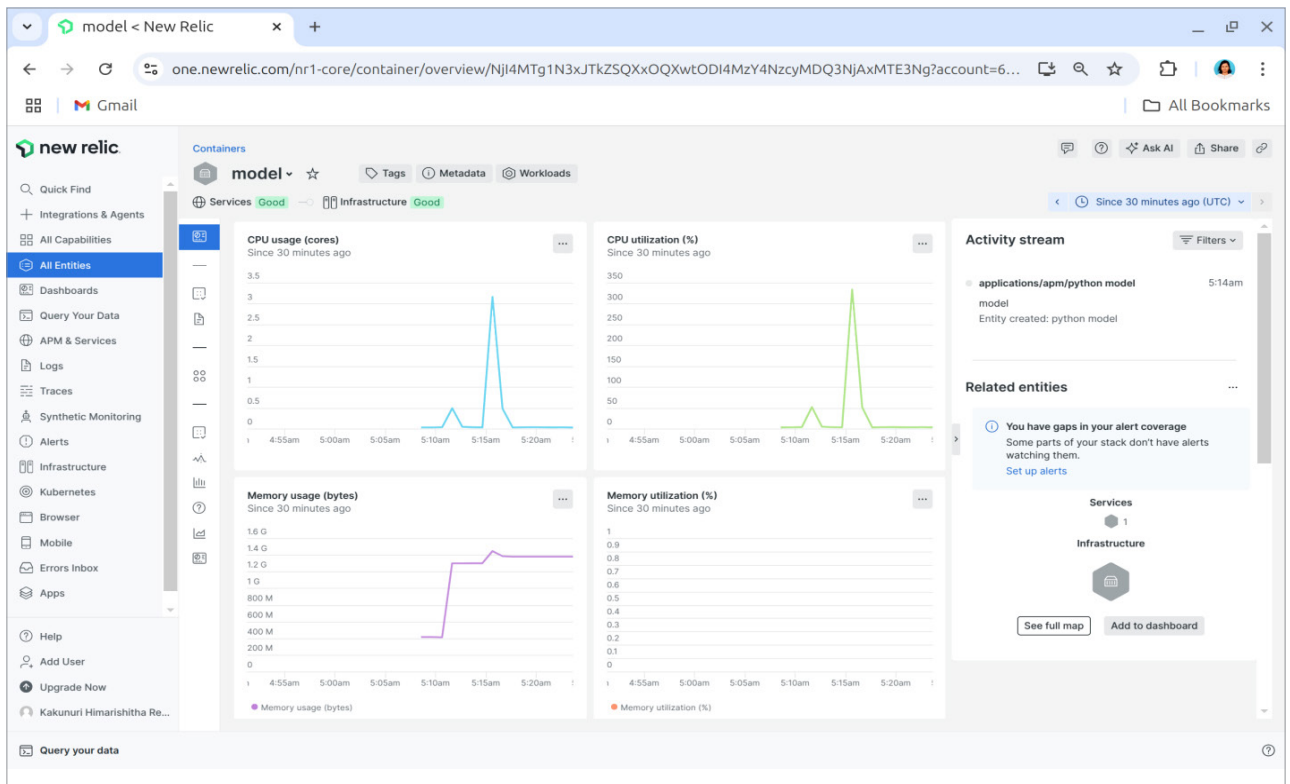


Figure 7: New Relic dashboard displaying performance metrics for a selected Docker container

- Once the agent is installed and configured, navigate to New Relic's infrastructure monitoring section to view all running Docker containers on the host as shown in Figure 6.

Container monitoring and metrics

After setting up the agent, New Relic automatically detects and monitors all running containers on the host machine. The dashboard in Figure 7 provides a comprehensive overview of container metrics including:

- CPU usage and memory consumption for each container
- Disk I/O and network activity metrics
- Container uptime and restart counts
- Logs and performance data for frontend, backend, and model services

This enables real-time monitoring of the containerised environment, facilitating proactive issue detection and performance optimisation.

The integration of New Relic significantly improves observability, performance management, and troubleshooting across the backend, frontend, machine learning model, and infrastructure. With real-time insights and proactive monitoring, the system ensures high availability and optimal performance.

New Relic effectively tracks backend response times, error rates, and database performance, while frontend monitoring analyses user interactions, page load times, and JavaScript errors. The machine learning model, hosted on Django and Docker, is continuously monitored for latency and resource usage, ensuring AI-driven components remain efficient. Infrastructure monitoring provides visibility into CPU, memory, disk, and network usage, helping prevent bottlenecks.

With historical performance analysis and proactive alerts, the system enables trend identification, rapid issue resolution, and enhanced stability.

As seen here, modern observability tools are extremely important for maintaining scalable, high-performance applications. Real-time monitoring, intelligent alerting, and historical analysis play a crucial role in efficient troubleshooting, system resilience, and better user experiences in containerised environments. **END** 🐧

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